

Assignment Q4

Deploying a WAR to a second EC2 instance (Tomcat) from Jenkins

Jenkins (JM) builds the WAR from the repo. The deployment target is a separate EC2 instance running Tomcat, not a Jenkins slave.

Prerequisite: Jenkins Server Setup

Do this on the Jenkins EC2 instance (Amazon Linux 2023) before starting the Q4 steps below.

Setup Step 1: Install Java

```
sudo yum install java-21-amazon-corretto.x86_64 -y
java -version
```

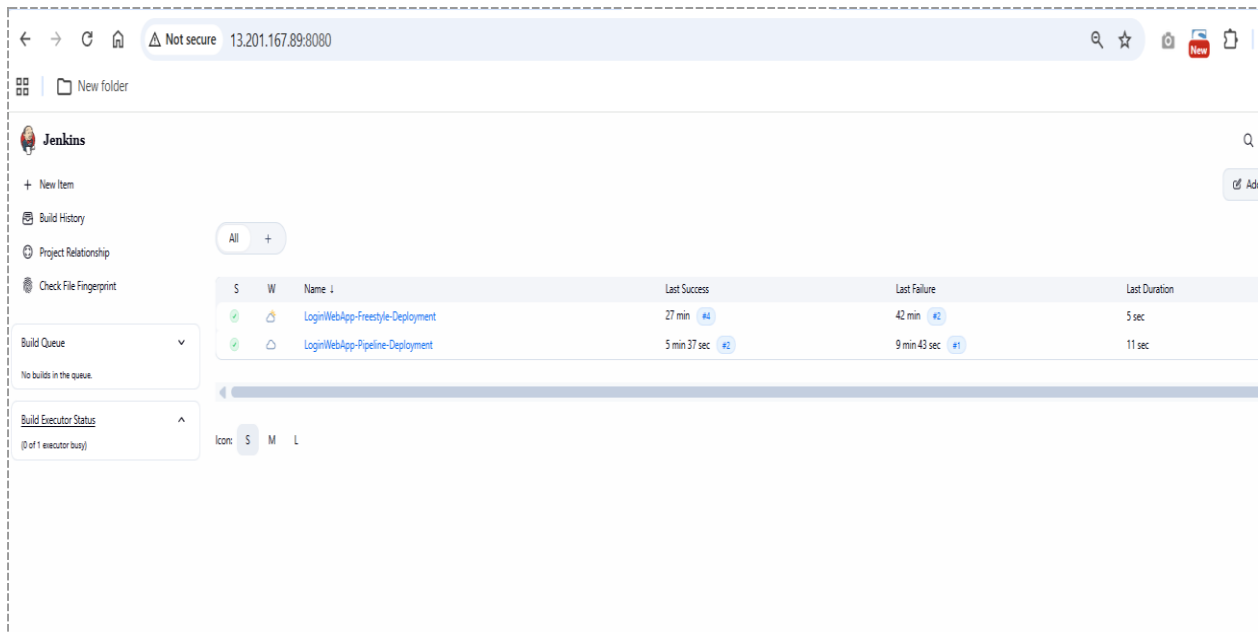
Setup Step 2: Install and start Jenkins

```
sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-
stable/jenkins.repo
sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
sudo yum install jenkins -y
sudo systemctl enable jenkins
sudo systemctl start jenkins
sudo systemctl status jenkins
```

Note: if Tomcat is already running on port 8080 on this same server, Jenkins will fail to start (port conflict). Move Jenkins to another port, for example 9090:

```
sudo mkdir -p /etc/systemd/system/jenkins.service.d
printf '[Service]\nEnvironment="JENKINS_PORT=9090"\n' | sudo tee
/etc/systemd/system/jenkins.service.d/override.conf
sudo systemctl daemon-reload
sudo systemctl reset-failed jenkins
sudo systemctl start jenkins
```

Also open the Jenkins port (8080, or 9090 if changed) in this instance's security group.

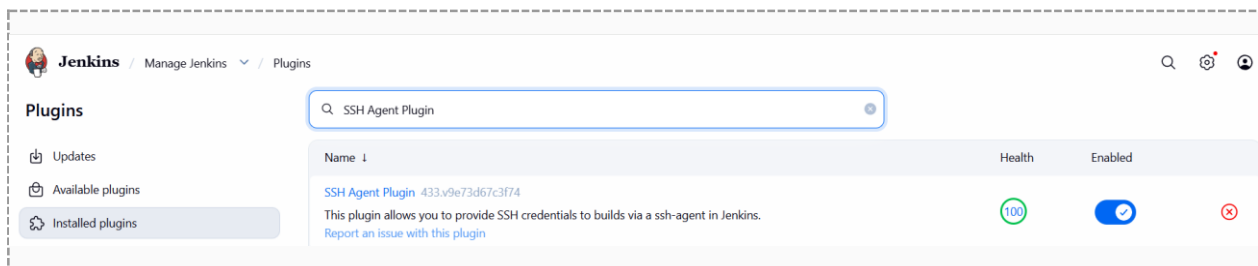


Setup Step 3: Unlock Jenkins and install plugins

- Open `http://<JENKINS-EC2-PUBLIC-IP>:8080` (or :9090 if you changed the port).
- Get the initial admin password:

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

- Choose Install suggested plugins, then create the admin user.
- Install the SSH Agent plugin: Manage Jenkins → Plugins → Available plugins → "SSH Agent" (needed for the sshagent deploy step in Q4).



Setup Step 4: Install and verify Git

Git must be available on the Jenkins VM so that Jenkins can check out the GitHub repository.

```
sudo yum install git -y
git --version
```

```
2026-10-07 16:53.31 /home/mobaxterm/Desktop/aws key git --version
git version 2.45.1
```

Setup Step 5: Install Maven

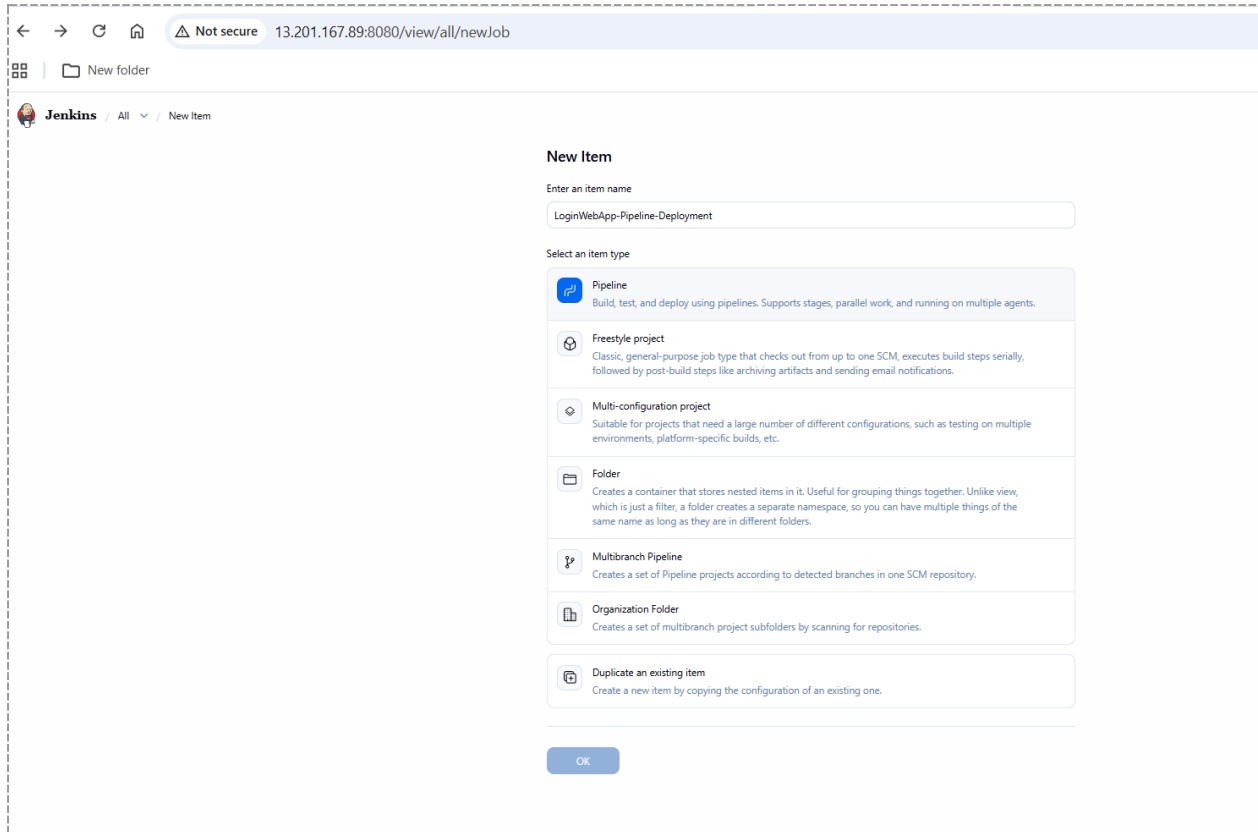
Needed to build the WAR (target/LoginWebApp.war) if the project is a Maven project.

```
sudo yum install maven -y
mvn -version
```

```
2026-10-07 17:00.09 /home/mobaxterm/Desktop/aws key mvn -version
MINGW support requires --add-opens java.base/java.lang=ALL-UNNAMED
MINGW support requires --add-opens java.base/java.lang=ALL-UNNAMED
Apache Maven 3.9.16 (2bdd9fddda4b155ebf8000e807eb73fd829a51d5)
Maven home: C:\Tools\apache-maven-3.9.16
Java version: 25.0.3, vendor: Eclipse Adoptium, runtime: C:\Users\darsh\AppData\Local\Programs\Eclipse Adoptium\jdk-25.0.3.9-hotspot
Default locale: en_IN, platform encoding: UTF-8
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"
```

Setup Step 6: Create the Jenkins Freestyle project

- On the Jenkins dashboard, click New Item.
- Enter a name, for example LoginWebApp-Freestyle-Deployment.
- Select Freestyle project and click OK.



Setup Step 8: Add the build step and run the job

Add pipeline script

```
pipeline {
  agent any

  stages {

    stage('Checkout') {
      steps {
        git branch: 'master',
            url: 'https://github.com/Shantanumajan6/project'
      }
    }

    stage('Fix WAR Plugin Temporarily') {
      steps {
        sh '''
          set -e

          echo "==== Adding compatible WAR plugin temporarily ====="

          sed -i '/<finalName>LoginWebApp<\/finalName>/a\\
          <plugins>\\
          <plugin>\\
            <groupId>org.apache.maven.plugins</groupId>\\
            <artifactId>maven-war-plugin</artifactId>\\
            <version>3.4.0</version>\\
          </plugin>\\
          </plugins>' pom.xml

          echo "==== WAR plugin configuration ====="
          grep -A8 -B2 "maven-war-plugin" pom.xml
        '''
      }
    }

    stage('Build WAR') {
      steps {
        sh '''
          set -e

          echo "==== Building WAR ====="

          mvn clean package
```

```

        echo "===== Generated WAR ====="
        ls -lh target/LoginWebApp.war
    ""
}
}

stage('Deploy to Tomcat EC2') {
    steps {
        sshagent(credentials: ['ec2-key']) {
            sh ""
            set -e

            echo "===== Deploying WAR to Tomcat ====="

            scp -o StrictHostKeyChecking=no \
                target/LoginWebApp.war \
                ec2-user@15.252.140.193:/opt/tomcat/webapps/LoginWebApp.war

            echo "===== Restarting Tomcat ====="

            ssh -o StrictHostKeyChecking=no \
                ec2-user@15.252.140.193 \
                "/opt/tomcat/bin/shutdown.sh || true; sleep 3; /opt/tomcat/bin/startup.sh"

            echo "===== Deployment completed ====="
            ""
        }
    }
}

post {
    success {
        echo '===== Pipeline completed successfully ====='
    }

    failure {
        echo '===== Pipeline failed ====='
    }
}
}

```

- Click Save, then Build Now.
- Open the build number → Console Output and confirm it ends with Finished: SUCCESS.

The deploy to the second EC2 is added later in Q4, Step 5.


```
wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.60/bin/apache-tomcat-10.1.60.tar.gz
sudo mkdir -p /opt/tomcat
sudo tar xzvf apache-tomcat-10.1.60.tar.gz -C /opt/tomcat --strip-components=1
sudo chmod +x /opt/tomcat/bin/*.sh
sudo chown -R ec2-user:ec2-user /opt/tomcat
```

The chown line lets ec2-user copy the WAR into /opt/tomcat/webapps when Jenkins deploys.

Start Tomcat:

```
/opt/tomcat/bin/startup.sh
```

```
[root@ip-172-31-8-29 ~]# /opt/tomcat/bin/startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-172-31-8-29 ~]#
```

Step 4: Verify Tomcat

Check from the instance:

```
curl -I http://localhost:8080/
```

Or from a browser:

```
http://<NEW-EC2-PUBLIC-IP>:8080/
```

```
[root@ip-172-31-8-29 ~]# curl -I http://localhost:8080/
HTTP/1.1 200
Content-Type: text/html; charset=UTF-8
Date: Wed, 07 Oct 2026 12:22:15 GMT
[root@ip-172-31-8-29 ~]#
```

Step 5: Configure Jenkins to deploy remotely

On your Jenkins EC2:

1. Add your .pem key: Manage Jenkins → Credentials → System → Global credentials → Add Credentials. Kind: SSH Username with private key, ID: ec2-key, Username: ec2-user, paste the contents of the .pem file.
2. Add the deploy step using Option A (Freestyle, as in Assignment 3) or Option B (Pipeline).

Option B: Pipeline project (Jenkinsfile deploy stage)

```
• stage('Deploy to Remote EC2') {  
•     steps {  
•         sshagent(['ec2-key']) {  
•             sh '''  
•                 scp -o StrictHostKeyChecking=no target/LoginWebApp.war ec2-  
•                 user@<NEW-EC2-IP>:/opt/tomcat/webapps/  
•                 ssh -o StrictHostKeyChecking=no ec2-user@<NEW-EC2-IP>  
•                 "/opt/tomcat/bin/shutdown.sh || true && /opt/tomcat/bin/startup.sh"  
•                 '''  
•             }  
•         }  
•     }  
• }
```

Jenkins / LoginWebApp-Pipeline-D... / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

☐ Poll SCM ?

☐ Trigger builds remotely (e.g., from scripts) ?

Pipeline
Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```
1 ~ pipeline {  
2   agent any  
3  
4   stages {  
5  
6     stage('Checkout') {  
7       steps {  
8         git branch: 'master',  
9         url: 'https://github.com/Shantanujana6/project'  
10      }  
11    }  
12  
13    stage('Fix WAR Plugin Temporarily') {  
14      steps {  
15        sh '''  
16          scp -o StrictHostKeyChecking=no target/LoginWebApp.war ec2-  
17          user@<NEW-EC2-IP>:/opt/tomcat/webapps/  
18          ssh -o StrictHostKeyChecking=no ec2-user@<NEW-EC2-IP>  
19          "/opt/tomcat/bin/shutdown.sh || true && /opt/tomcat/bin/startup.sh"  
20          '''  
21      }  
22    }  
23  }  
24 }
```

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Advanced

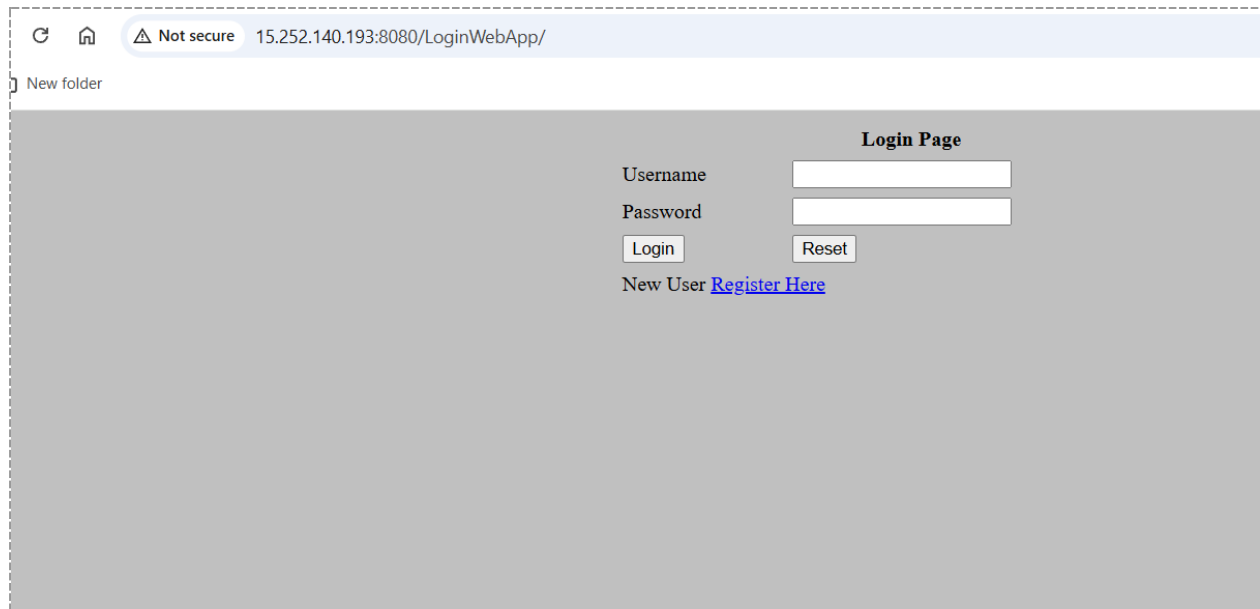
Advanced ▾

Save Apply

Step 6: Test deployment

After the pipeline runs, open:

<http://15.252.140.193:8080/LoginWebApp/>



Summary

You do not need a Jenkins slave. Launch a second EC2 instance, install Java and Tomcat, and let Jenkins deploy the WAR remotely via SSH/SCP using your existing .pem key.